Steven Flynn

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Research Interests

Hypoelliptic and Dispersive PDEs. Spectral Theory. Microlocal Analysis. Analysis and Representation Theory on Lie groups. Geometric Inverse Problems. Sub-Riemannian geometry. Mathematical foundations of AI interpretability.

Employment

2024–present	**Postdoctoral Fellow**, University College London
2023–2024	**Postdoctoral Fellow**, University of Padova
2021–2023	**Postdoctoral Fellow**, University of Bath
2014–2020	**Graduate Student Instructor & Graduate Teaching Assistant**, University of California, Santa Cruz

Fellowships and Grants

Clay Mathematics Institute Enhancement and Partnership Program	2022
Isaac Newton Institute Fellowship	2021
UCSC Mathematics Summer Research Fellowship	2020
UCSC Mathematics Year-End Fellowship	2019
UCSC Travel/Research Grants	2015–2020
UCSC Early Academic Achievement Award	2015

Education

2014–2020	**Ph.D. Mathematics**, University of California, Santa Cruz (UCSC) *Thesis: Unraveling Geodesic X-ray Transforms on the Heisenberg Group* *Co-advisors:* François Monard, Richard Montgomery
2014–2015	**M.A. Mathematics**, UCSC
2012–2014	**B.A. Mathematics**, UCSC

Research Papers

Journal Articles

[1] Clotilde Fermanian-Kammerer, Véronique Fischer, and Steven Flynn. Quantization on filtered manifolds. arXiv:2412.17448, 2024. To appear in *Memoirs of the European Mathematical Society*

[2] Clotilde Fermanian-Kammerer, Véronique Fischer, and Steven Flynn. Geometric invariance of the semi-classical calculus on nilpotent graded Lie groups. *Journal of Geometric Analysis*, 2023.

[3] Steven Flynn. Injectivity of the Heisenberg X-ray transform. *Journal of Functional Analysis*, 280.5 (2021): 108886.

Refereed Proceedings and Volumes

[4] Steven Flynn. Singular Value Decomposition for the X-Ray Transforms on the Reduced Heisenberg Group, and a Two-Radius Theorem. In *Extended Abstracts 2021/2022: Methusalem Lectures*, eds. D. Cardona, J. Restrepo and M. Ruzhansky, Trends in Mathematics 3, Birkhäuser, 2024, pp. 15–26.

[5] Clotilde Fermanian-Kammerer, Véronique Fischer, and Steven Flynn. Some Remarks on Semi-Classical Analysis on Two-Step Nilmanifolds. In *Quantum Mathematics I*, Springer INdAM Series 57, 2023, pp. 129–162.

Preprints

[6] Davide Barilari and Steven Flynn. Refined Strichartz estimates for sub-Laplacians in Heisenberg and H-type groups. arXiv:2501.04415, 2025. Submitted to *Journal of Differential Equations*.

[7] Steven Flynn. Injectivity and Stability for the sub-Riemannian X-ray Transform on H-type Groups. arXiv:2312.00594, 2023. Submitted to *Analysis & PDE*.

Talks

Strichartz Estimates on H-type groups

May 2025	UC Santa Cruz
Jan 2025	Lund University
Dec 2024	University College London

A Microlocal Calculus on Filtered Manifolds

Dec 2025	HADES Seminar, UC Berkeley
Dec 2025	Geometric Structures Seminar, SISSA
Jan 2025	University College London
May 2024	Dispersion and Geometry in Padova

The sub-Riemannian X-ray Transform on H-type groups

Nov 2023	Junior Methusalem Seminar, Ghent University
Oct 2023	University College London
Nov 2022	Geometry and Analysis Seminar, University of Bristol

Remarks on the Semi-classical analysis of Step 2 Nilpotent groups

May 2023	Operator Algebras in the South of the UK, University of Southampton
Mar 2023	CAGE mini-seminar, Sorbonne Université

Geometric Invariance of the Semi-Classical Calculus on Graded Lie groups

Nov 2022	Conference on Noncommutative Analysis and PDEs, Queen Mary University

The Heisenberg X-ray transform

May 2022	Analysis and Differential Geometry International Seminar, University of Aveiro
Mar 2022	AGeNT Seminar, University of Bath
Nov 2021	Problèmes Spectraux en Physique Mathématique, Institut Henri Poincaré
Sep 2021	Bath Analysis Seminar, University of Bath
Apr 2021	Sub-Riemannian Seminars (online)
May 2020	UCSC Geometry and Analysis Seminar

Integral Geometry on Contact Manifolds

Jan 2020	Joint Math Meeting, Denver, Colorado
Nov 2019	Mathematical Sciences Research Institute, Berkeley, CA
Sep 2019	MSRI, Berkeley, CA

Service

Reviewer, *Journal of Functional Analysis*	2022–2023
Organizer, Bath Analysis Seminar	2022
Organizer, University of Bath Postdoc Away Day	2022
Organizer, Graduate Student Seminar, MSRI	2019
Organizer, Microlocal Analysis Seminar, UCSC	2018
Organizer, Graduate Differential Geometry Seminar, UCSC	2018

Professional Activities

Participant, MPS Workshop on Lean	2025
Program Associate, MSRI Microlocal Analysis Program	2019
Mentor, UCSC Directed Reading Program	2018

Computing

Python, MATLAB, Lean 4, Git.

Teaching

Taught Course Centre (Bath, Bristol, Imperial, Oxford, Swansea, Warwick)

Autumn 2026	Analysis of the Sub-Laplacian (Term 1)

University College London

Autumn 2025	MATH0071: Spectral Theory

Supervision

2026–27	MSci project: degenerate diffusions and hypoellipticity, UCL
Aug–Oct 2026	Summer research project: equidistribution and ergodic theorems, UCL
2018	Mentor, Directed Reading Program, UCSC

University of California, Santa Cruz

Summer 2019	Math 105A: Real Analysis
Spring 2019	Math 105B: Real Analysis II
Summer 2018	Math 105A: Real Analysis
Summer 2017	Math 3: Precalculus

Graduate Teaching Assistant

Six years of continuous teaching experience (20 hours/week). Duties included delivering lectures and review sessions, grading homework and examinations, and holding office hours.

Courses: calculus, vector calculus, linear algebra, real analysis, introduction to proofs, programming for mathematics.